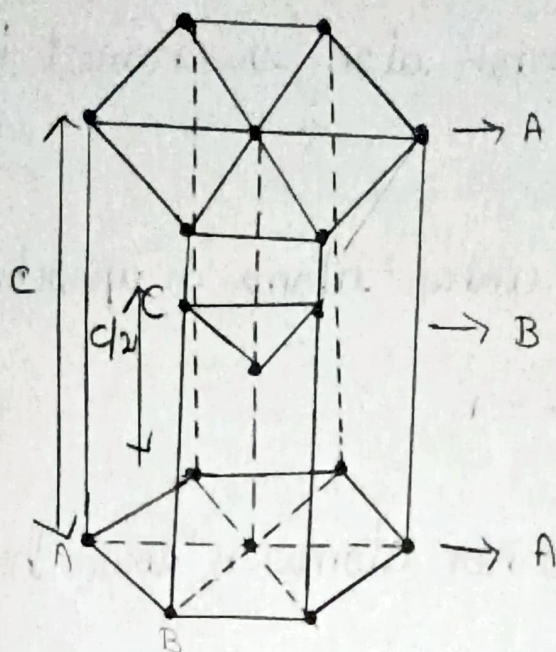


## Hexagonal Close-Packed Structure



It has three layers, At bottom layer,

it has 6 corners and each corner has 1 atom.

One more atom present at the centre of bottom face. At top layer, it has 6 corners, and each corner has 1 atom and one more atom present at centre of top face. Then the middle layer has 3 atoms at a distance  $\frac{c}{2}$  from top or bottom layers. There are 3 atoms inside the unit cell. Height of the hexagon is  $c$ .



No. of atoms per unit cell

Each corner atom is shared by other unit cell.

No. of corner atoms in upper hexagonal plane

$$= \frac{1}{6} \times 6 = 1$$

No. of corner atoms in lower hexagonal plane

$$= \frac{1}{6} \times 6 = 1$$

Each face centered atom is shared by 2 unit cells

Total No. of face centered atoms in the

$$\text{lower and upper planes} = \frac{1}{2} \times 2 = 1$$

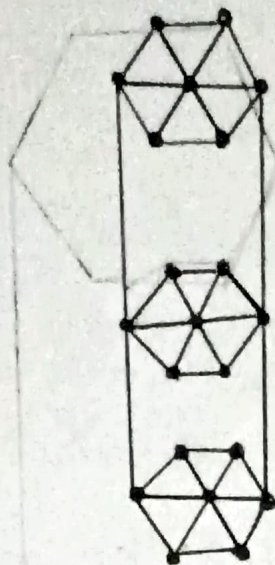
There are 3 atoms present in middle layer which is not shared

Total No. of atoms in HCP.

$$= 1 + 1 + 1 + 3 = 6$$



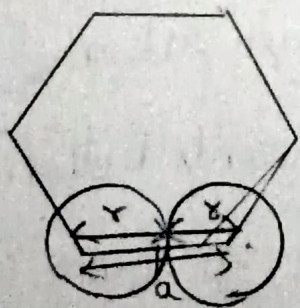
## Coordination Number



In bottom layer, face centered atom has 6 nearest neighbors. At a distance  $\frac{c}{2}$ , 3 atoms above and below layer.  $\therefore$  There are 12 neighbors.

Coordination No = 12

## Atomic radius



Let  $a$  be the side and  $r$  be the radius,

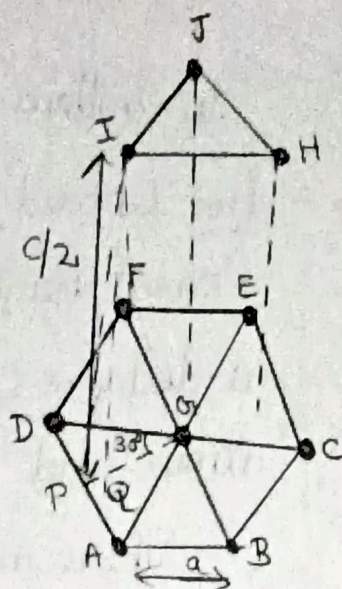
From the diagram,

$$2r = a$$

$$r = \frac{a}{2}$$



## Calculation of $\frac{c}{a}$ ratio



Let ' $c$ ' be the height of the unit cell  
 and ' $a$ ' be the distance between neighbouring atoms.  
 Let A, B, C, D, E and F be corner atoms, G  
 be the base atom of bottom layer.  
 Let H, I and J be middle layer atoms.

Here A, D, and G represents bottom  
 layer atoms. Let draw normal PG from G  
 to line AD. It bisects line AD at point P.

From  $\triangle DPG$ ,

$$\cos 30^\circ = \frac{PG}{DG}$$



$$PGI = DG \cos 30^\circ$$

$$PGI = DG \times \frac{\sqrt{3}}{2}$$

$$PGI = a \frac{\sqrt{3}}{2} \rightarrow (1)$$

From figure,

$$GIQ = \frac{2}{3} PGI \rightarrow (2)$$

Sub (1) in (2)

$$GIQ = \frac{2}{3} \times \frac{\sqrt{3}}{2} \times a$$

$$GIQ = \frac{a}{\sqrt{3}} \rightarrow (3)$$

From  $\triangle QGI$ ,

$$IG^2 = IQ^2 + QG^2$$

$$\left(\frac{C}{2}\right)^2 =$$

$$a^2 = \left(\frac{C}{2}\right)^2 + \frac{a^2}{3}$$

$$\left(\frac{C}{2}\right)^2 = a^2 - \frac{a^2}{3}$$

$$= \frac{3a^2 - a^2}{3}$$

$$= \frac{2a^2}{3}$$

$$\frac{C}{2} = \frac{\sqrt{2}a}{\sqrt{3}}$$



$$\frac{c}{a} = \frac{2 \times \sqrt{3}}{\sqrt{3}}$$

$$\frac{c}{a} = \frac{\sqrt{8}}{\sqrt{3}}$$

$$\frac{c}{a} = 1.633$$

Packing factor

Volume of all atoms in unit cell

$$V = 6 \times \frac{4}{3} \pi r^3$$

$$V = 6 \times \frac{4}{3} \pi \times \frac{a^3}{8}$$

$$V = \pi a^3 \rightarrow (1)$$

Volume of unit cell  $V$

Volume of hexagon = Area of base  $\times$  Height of triangle

Base of hexagon has 6 similar triangles

$$= \frac{1}{2} \times a \times \frac{a\sqrt{3}}{2} \times 6 \times c$$

$$= \frac{a^2 \sqrt{3}}{4} \times 6 \times c$$

$$= \frac{3\sqrt{3}a^2 c}{2} \rightarrow (2)$$



$$\text{Packing factor P.F} = \frac{v}{V}$$

$$= \frac{2\pi a^3}{3\sqrt{3}a^2c}$$

$$= \frac{2\pi a}{3\sqrt{3}c}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{8}}$$

$$= \frac{2\pi}{3 \times \sqrt{8}}$$

$$= \frac{2\pi}{3 \times 2\sqrt{2}}$$

$$= \frac{3.14}{3 \times 1.414}$$

$$= 0.74$$

$$\text{P.F} = 74\%$$

Example: Magnesium, Titanium  
Chromium